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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech. (EE) (2012 Onwards) / (Electrical & Electronics Engg.) (2011 Onwards)
B.Tech. (Electronics & Electrical Engg./ Electrical Engineering & Industrial Control) (2012 to 2017) (Sem.–3)

ELECTRONIC DEVICES AND CIRCUITS

Subject Code : BTEE-304
M.Code : 57095

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTION TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

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SECTION-A

Answer briefly :

Q1 How do you distinguish between tow leads of an LED?

Q2 Why the space charge region is also called the depletion region?

Q3 How transistor "α" and "β" are related?

Q4 Why is the input impedance of the MOSFET very high?

Q5 List the ideal characteristics of operational amplifier.

Q6 Define CMRR.

Q7 Why the series regulator is mostly commonly used regulator circuitry?

Q8 Define line and load regulation.

Q9 State the conditions under which a feedback amplifier works as an oscillator?

Q10 What is the basic principle of operation of crystal oscillator?

SECTION-B

Q11 Draw the circuit diagram of half wave rectifier. What is the minimum frequency of ripple in its output?

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Q12 Draw CB configuration circuit diagram and explain with characteristics.

Q13 Explain the working of RC phase shift oscillator and state some of its applications.

Q14 In the circuit shown below $R_1 = 5k\Omega$ and $R_2 = 105\ k\Omega$.

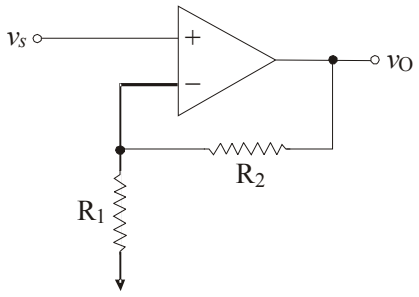

Fig.1

Assume that the op-amp is ideal. Determine the closed-loop gain, $G = v_0 / v_s$.

Q15 Draw the pin package and circuit diagram of LM317 and explain its working.

SECTION-C

Q16 Draw the various types of biasing circuits and explain voltage divider biasing circuit using expressions.

Q17 If R_S is connected in series with the zener diode to limit the current flow through the diode with the voltage source. A 10.0V stabilized power supply is required to be produced from a 24V DC power supply input source. The maximum power rating P_z of the Zener diode is 2W. Using the above data calculate :

a) The maximum current flowing through the Zener diode.

b) The minimum value of the series resistor, R_s

c) The Zener current I_z at full load.

d) The load current I_L if a load resistor of $2k\Omega$ is connected across the Zener diode.

Q18 Explain **any two** with necessary diagrams ;

a) 555 timer.

b) SMPS.

c) RC low pass filter.

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